

Shu Hu

PhD Student in Microbiology · Fudan University

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[Google Scholar](#) · [ResearchGate](#) · [GitHub](#) · [ORCID](#)

Research profile

I am a PhD student in the Department of Microbiology, School of Life Sciences, Fudan University, with an expected graduation date of June 2029. My research interests span viral evolution, genomic epidemiology and computational approaches to emerging infectious diseases. I combine research experience at the University of Hong Kong with industry experience in bioinformatics, including transcriptomics, single-cell analysis and genomic data analysis workflows.

Education

Fudan University | PhD student in Microbiology

Sep 2025 – Jun 2029 (expected)

Department of Microbiology, School of Life Sciences

The University of Edinburgh | MSc in Bioinformatics, with Merit

Sep 2021 – Nov 2022

Training in genomics, bioinformatics algorithms, statistical analysis and scientific programming.

Sichuan Agricultural University | BSc in Biological Science

Sep 2016 – Jun 2020

GPA: 3.8/5.0; average mark: 88/100.

Professional experience

Shanghai Rightongene Technology Co., Ltd. | Bioinformatics Engineer

2024 – May 2025; Sep 2022 – May 2023

- Developed analysis methods for BCL6, BCL2 and MYC gene fusions in DLBCL sequencing workflows, addressing discrepancies between NGS and FISH results.
- Established low-depth whole-genome sequencing workflows for copy-number analysis using QDNAseq, Control-FREEC and ichorCNA.
- Built upstream and downstream single-cell RNA-seq workflows and collaborated with IT colleagues to deploy analyses on an automated platform.
- Developed RNA-seq expression-based models for leukemia molecular subtyping.

The University of Hong Kong | Research Assistant I

2023–2024

- Conducted bioinformatics analyses of viral evolution and genomic epidemiology, with a focus on influenza and RSV.
- Integrated genomic, clinical and immunological datasets and contributed to collaborative research publications.

Sichuan WoWente Biotechnology Co., Ltd. | Reagent Developer

May 2020 – May 2021

- Contributed to sex-hormone immunoassay reagent development and an inhibin A (INH-A) project.

Publications

Name in bold. Published versions are listed once; earlier preprints of the same work are omitted.

1. Ye D, Hu F, Hu H, **Hu S**, Tan Y, Ouyang W, Li SZ, Cui J, Dong N. ViroBench: Benchmarking Nucleotide Foundation Models on Viral Genomics Tasks. *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. 2026; 12679–12690. doi:10.1145/3770855.3819057 [Conference paper]
2. Siegers JY, Xie R, Edwards KM, Byrne AMP, **Hu S**, et al. Resurgence of Zoonotic Highly Pathogenic Avian Influenza A(H5N1) Virus in Cambodia. *New England Journal of Medicine*. 2025; 393(16):1650–1652. doi:10.1056/NEJMc2504302 [Correspondence]
3. **Hu S**, Cui J. Cattle H5N1 outbreak in US driven by cow's "cross-nursing" behaviour: elucidating a novel transmission mechanism. *Emerging Microbes & Infections*. 2025; 14(1):2547736. doi:10.1080/22221751.2025.2547736 [First author]
4. Yang K, Nizami S, **Hu S**, Zou L, Deng H, Xie J, Guo Q, Edwards KM, Dhanasekaran V, Yen HL, Wu J. Genetic diversity of highly pathogenic avian influenza H5N6 and H5N8 viruses in poultry markets in Guangdong, China, 2020–2022. *Journal of Virology*. 2025; 99(1):e01145-24. doi:10.1128/jvi.01145-24 [Research article]
5. Xie R, Adam DC, **Hu S**, Cowling BJ, Gascuel O, Zhukova A, Dhanasekaran V. Integrating Contact Tracing Data to Enhance Outbreak Phylodynamic Inference: A Deep Learning Approach. *Molecular Biology and Evolution*. 2024; 41(11):msae232. doi:10.1093/molbev/msae232 [Research article]

Preprints

The following preprints have not been peer reviewed.

1. Schwab LSU, Xie R, Reilly E, Aban M, Spirason N, Deng YM, **Hu S**, et al. Epistasis facilitates the long-term antigenic evolution of the influenza B virus hemagglutinin. *bioRxiv*. 2026; Posted 22 July 2026. doi:10.64898/2026.07.22.739994 [Preprint]
2. Xie R, Zhukova A, Peña PG, Iglesias G, **Hu S**, Wang J, Tsang TK, Dhanasekaran V, Kraemer MUG, Pybus OG, Gascuel O. Scalable deep-learning-based inference of time-varying transmission dynamics from outbreak phylogenies. *medRxiv*. 2026; Posted 10 May 2026. doi:10.64898/2026.05.07.26352673 [Preprint]

Other publication

1. **Hu S**, Zeng L, Luo L, et al. Research progress on the regulation of ornamental fish coloration. *Scientific Fish Farming*. 2019; (5):77–78. doi:10.14184/j.cnki.issn1004-843x.2019.05.039 [In Chinese · first author]

Selected research and software projects

Computational phylodynamics | Research initiated in 2023

Contributed to collaborative research combining phylogenetic and epidemiological information for outbreak analysis. Related work was published in Molecular Biology and Evolution in 2024, with a further preprint in 2026.

[Publication](#)

Single-cell RNA-seq analysis workflows | Oct 2022 – Mar 2023

Developed Shell-based upstream processing and R-based downstream analysis with configurable parameters for automated single-cell RNA-seq workflows.

[GitHub](#)

RNA-seq models for leukemia subtyping | Oct 2022 – Mar 2023

Built expression-based models to classify leukemia samples into molecular subtypes and support the interpretation of transcriptomic data.

[GitHub](#)

Interactive transcriptomics with R Shiny | Feb – Sep 2022

Developed reusable Shiny modules and R scripts for interactive exploration of RNA-seq datasets during MSc study at the University of Edinburgh.

[Related code](#)

Undergraduate research on fish pigmentation | Sep 2019 – Sep 2020

Led a student project investigating ASIP expression across tissues in neon tetra; coordinated project timelines and laboratory work.

Technical skills

Programming and computing: R, Python, Linux, Shell, Docker, SQL

Genomic data analysis: RNA-seq, single-cell RNA-seq, whole-genome sequencing, copy-number analysis and workflow development

Evolution and modelling: Phylogenetics, BEAST, Nextstrain, statistical analysis and deep learning

Visualisation and tools: R Shiny, interactive data exploration and scientific visualisation

Honours and training

- 2019 Second Prize, 8th Biology Knowledge and Experimental Skills Competition
- 2019 Excellence Award, 3rd National Undergraduate Environmental Knowledge Competition
- 2018 Outstanding Team and Advanced Individual, Three to Countryside Social Practice
- 2018 Start Your Business training certificate

Academic and community activities

Attended the Kunming Institute of Botany, Chinese Academy of Sciences, summer camp (2019). Led the New Era social practice team, coordinating activities, communications and reporting (2018).